Mathematics practice booklet: Grade 9 expectations

Grade 9 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 9 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

Section 1: Patterns, equations and algebra thinking

Standards practised: A-SSE.A.1, A-SSE.B.3, A-APR.A.1, A-CED.A.1, A-CED.A.4, A-REI.A.1, A-REI.B.3, A-REI.B.4, A-REI.C.6, F-IF.A.2, F-LE.A.1

A worked example

Solve x^2 + 2x - 15 = 0 by factoring. Then check one of your answers.

1. Look for two numbers that multiply to -15 and add to 2: they are 5 and -3.
2. So x^2 + 2x - 15 = (x + 5)(x - 3) = 0.
3. A product is 0 only when one factor is 0: x + 5 = 0 or x - 3 = 0.
4. So x = -5 or x = 3.
5. Check x = 3: 9 + 6 - 15 = 0. It works.

x = -5 or x = 3

Questions

1. Solve: 4x - 9 = 19

2. Solve: 3(2x - 1) = 5x + 4

3. Solve the inequality: 2x + 5 > 17

4. Solve the inequality: -3x <= 12. What happens to the inequality sign, and why?

5. The perimeter of a rectangle is P = 2l + 2w. Solve the formula for w. Then find w when P = 30 and l = 9.

6. Solve the system: y = 3x - 2 and 2x + y = 13

7. Solve the system by elimination: 3x + 2y = 16 and x - 2y = 0

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8. If f(x) = 2x^2 - 3x + 1, find f(3).

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9. Expand (x + 3)(x - 5). Then factor x^2 + 7x + 12.

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10. Solve by factoring: x^2 - 5x + 6 = 0

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11. Noura has AED 50. Karak costs AED 3 a cup and she also buys a snack for AED 8. Write an inequality for the number of cups c she can buy, and solve it.

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12. Maryam's savings start at AED 200 and grow by AED 50 each month. Her cousin Hamdan starts with AED 100 and his savings double each month. Who has more after 4 months? Which one is linear and which is exponential?

Section 2: Shapes, measuring and space

Standards practised: G-GPE.B.5, G-GPE.B.7, G-GMD.A.3

A worked example

Line A is y = 2x + 3. Write the equation of the line through (4, 1) that is perpendicular to line A.

1. Line A has slope 2.
2. A perpendicular line has the negative reciprocal slope: -1/2.
3. Use y - 1 = -1/2 (x - 4), so y - 1 = -1/2 x + 2.
4. Add 1 to both sides: y = -1/2 x + 3.
5. Check: at x = 4, y = -2 + 3 = 1. The line passes through (4, 1).

y = -1/2 x + 3

Questions

1. Find the distance between (1, 2) and (7, 10).

2. Find the midpoint of the segment from (-2, 4) to (6, 10).

3. The midpoint of a segment is (3, 5) and one endpoint is (1, 2). Find the other endpoint.

4. Find the slope of the line through (2, 3) and (6, 11).

5. Write the equation of the line through (0, 5) that is parallel to y = 2x + 1.

6. A line has slope 3. What is the slope of any line perpendicular to it?

7. Are the lines y = 2/3 x + 1 and y = -3/2 x + 4 perpendicular? Explain.

8. Write the equation of the line through (1, 4) that is perpendicular to y = -1/2 x + 3.

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9. A triangle has vertices (0, 0), (4, 0) and (4, 3). Find its perimeter.

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10. A cylindrical water tank has radius 7 m and height 10 m. Find its volume using 22/7 for pi.

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11. A cone has radius 6 cm and height 10 cm. Find its volume using 3.14 for pi.

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12. A ball has radius 3 cm. Find its volume in terms of pi, and then using 3.14. If the radius doubles, how many times as large does the volume become?

Section 3: Data, graphs and chance

Standards practised: S-ID.A.1, S-ID.A.2, S-ID.A.3, S-ID.B.5, S-ID.B.6, S-ID.C.7, S-ID.C.8, S-ID.C.9

A worked example

Find the mean and the standard deviation of the data: 4, 6, 8, 10, 12.

1. Mean: (4 + 6 + 8 + 10 + 12) / 5 = 40 / 5 = 8.
2. Distances from the mean: -4, -2, 0, 2, 4. Square them: 16, 4, 0, 4, 16.
3. Add the squares: 40. Divide by 5: 8. This is the variance.
4. The standard deviation is the square root of 8, about 2.8.

Mean 8; standard deviation about 2.8

Questions

1. A survey of 60 Grade 9 students asked how they get to school. Girls: 12 walk and 18 take the bus. Boys: 15 walk and 15 take the bus. How many students walk in all?

2. In the same survey, what percentage of the girls walk? What percentage of the boys walk?

3. In the same survey, what percentage of ALL the students are boys who take the bus?

4. A line of fit for the exam score y against the hours of study x is y = 5x + 50. Predict the score for 4 hours of study.

5. In y = 5x + 50, explain what the slope 5 and the y-intercept 50 mean.

6. A line of fit predicts a score of 70 but the real score was 74. Find the residual (actual minus predicted).

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7. Which correlation coefficient shows the strongest negative relationship: r = 0.9, r = -0.8, r = 0.1 or r = -0.2?

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8. Children with bigger shoe sizes tend to read better. Does a bigger shoe size cause better reading? Explain.

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9. Data: 2, 4, 5, 7, 8, 9, 12. Find the median, the lower quartile, the upper quartile and the interquartile range.

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10. For the same data, which five numbers mark the ends and the box of a box plot?

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11. Daily visitors to a small shop: 5, 6, 7, 8, 9. On a festival day 40 visitors came and this number was added to the list. Find the mean and the median before and after. Which is a better description of a usual day after adding it?

Section 4: Numbers and calculating

Standards practised: N-RN.A.1, N-RN.A.2, N-RN.B.3, N-Q.A.1

A worked example

Evaluate 8^(2/3) and then write 5^(1/2) as a radical.

1. The exponent 2/3 means: take the cube root, then square. The cube root of 8 is 2.
2. Square it: 2^2 = 4, so 8^(2/3) = 4.
3. An exponent of 1/2 means the square root, so 5^(1/2) is the square root of 5.

4 and the square root of 5

Questions

1. Simplify and evaluate: 2^5 x 2^3

2. Simplify and evaluate: 5^6 / 5^4

3. Evaluate: 4^(1/2), 27^(1/3) and 16^(3/2).

4. Write the cube root of 5 using a rational exponent.

5. Evaluate: 5^-2 and 7^0. Explain why 7^0 = 1.

6. Is 1/2 + 3/4 rational or irrational? Is 3 + the square root of 2 rational or irrational?

7. Is the product of the square root of 2 and the square root of 8 rational or irrational? Show why.

8. A falcon flies at 108 km per hour. Convert this to metres per second.

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9. A family car uses 8 litres of fuel for every 100 km. How many litres does it use on a 140 km trip?

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10. Write 5.2 x 10^-3 as an ordinary number, and write 45,000,000 in scientific notation.

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11. Calculate (2 x 10^4) x (3 x 10^3) and (6 x 10^8) / (3 x 10^5). Give each answer in scientific notation.

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12. A desalination plant makes 3 x 10^6 litres of fresh water a day. How many litres does it make in 7 days? Write the answer in scientific notation, and say which unit you would use to talk about it more easily.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. x = 7 (4x = 28)
2. x = 7 (6x - 3 = 5x + 4, so x = 7)
3. x > 6 (2x > 12)
4. x >= -4. The sign turns around because we divided both sides by a negative number.
5. w = (P - 2l) / 2. When P = 30 and l = 9, w = (30 - 18) / 2 = 6.
6. x = 3, y = 7 (2x + 3x - 2 = 13 gives 5x = 15)
7. x = 4, y = 2 (adding the equations gives 4x = 16; check: 12 + 4 = 16)
8. 10 (2 x 9 - 9 + 1)
9. x^2 - 2x - 15. And x^2 + 7x + 12 = (x + 3)(x + 4).
10. x = 2 or x = 3 ((x - 2)(x - 3) = 0)
11. 3c + 8 <= 50, so c <= 14. She can buy at most 14 cups.
12. Maryam: 200 + 4 x 50 = AED 400. Hamdan: 100 x 2^4 = AED 1,600. Hamdan has more. Maryam's is linear (adds the same amount); Hamdan's is exponential (multiplies by the same factor).

Section 2: Shapes, measuring and space

1. 10 units (changes of 6 and 8; 36 + 64 = 100)
2. (2, 7)
3. (5, 8) (3 is 2 more than 1, so 3 + 2 = 5; 5 is 3 more than 2, so 5 + 3 = 8)
4. 2 (rise 8 over run 4)
5. y = 2x + 5 (parallel lines have the same slope)
6. -1/3 (the negative reciprocal)
7. Yes. The slopes multiply to 2/3 x (-3/2) = -1.
8. y = 2x + 2 (slope 2; y - 4 = 2(x - 1))
9. 12 units (4 + 3 + 5, and the long side is the distance from (0, 0) to (4, 3): 16 + 9 = 25)
10. 1,540 cubic m (22/7 x 49 x 10)
11. 376.8 cubic cm (1/3 x 3.14 x 36 x 10)
12. 36 pi, about 113.04 cubic cm (4/3 x pi x 27). The volume becomes 8 times as large, because 2 cubed is 8.

Section 3: Data, graphs and chance

1. 27 (12 + 15)
2. Girls: 40% (12 out of 30). Boys: 50% (15 out of 30).
3. 25% (15 out of 60)
4. 70 (5 x 4 + 50)
5. Slope: each extra hour of study goes with about 5 more points. Intercept: with 0 hours of study the predicted score is 50.
6. 4 (74 - 70). The point lies 4 above the line.
7. r = -0.8: it is negative and closest to -1.
8. No. Age explains both: older children have bigger feet and have had more practice reading. Correlation does not mean causation.

9. Median 7; lower quartile 4; upper quartile 9; IQR = 9 - 4 = 5.
10. Minimum 2, lower quartile 4, median 7, upper quartile 9, maximum 12.
11. Before: mean 7, median 7. After: mean 12.5, median 7.5. The median is better because the value 40 pulls the mean up a lot.

Section 4: Numbers and calculating

1. 2^8 = 256 (add the exponents)
2. 5^2 = 25 (subtract the exponents)
3. 2, 3 and 64. (16^(3/2): the square root is 4, and 4 cubed is 64.)
4. 5^(1/3)
5. 1/25 and 1. Dividing 7^3 by 7^3 gives 7^0, and any number divided by itself is 1.
6. 1/2 + 3/4 = 5/4, which is rational. 3 + the square root of 2 is irrational.
7. Rational: the square root of 2 x the square root of 8 = the square root of 16 = 4.
8. 30 metres per second (108,000 m / 3,600 s)
9. 11.2 litres (8 x 1.4)
10. 0.0052 and 4.5 x 10^7
11. 6 x 10^7 and 2 x 10^3
12. 2.1 x 10^7 litres (21,000,000). Cubic metres or megalitres would be easier: 21,000 cubic metres.